

Logistics_Sensor_Gateway_Technical_Specification

Here is the Technical Requirement Document (TRD) based on the provided Functional Requirement Document (FRD).

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TR-GW-001: Ingest Raw Sensor Data

1. **Technical Requirement ID:** TR-GW-001

2. **Related Functional Requirement(s):** FR-GW-001

3. **Objective:** Implement a module to establish connections with IoT sensors, continuously receive raw text data streams, and temporarily buffer this incoming sensor data within the gateway.

4. **Target Cluster Configuration:**

- Cluster Name: Not specified in FRD
- Databricks Runtime Version: Not specified in FRD
- Node Type: Not specified in FRD
- Driver Node: Not specified in FRD
- Worker Nodes: Not specified in FRD
- Autoscaling: Not specified in FRD
- Auto Termination: Not specified in FRD
- Libraries Installed: Not specified in FRD

5. **Source Data Details:**

- Dataset Name: Raw IoT Sensor Data
- Location (Path/Table): Direct network connection from various IoT sensors
- Format: Raw text data streams
- Description: Contains location, fuel levels, and engine health information from delivery trucks.

6. **Target Data Details:**

- Output Name: Buffered Raw Sensor Text
- Location: Gateway's internal temporary storage/buffer
- Format: Raw text data
- Description: Temporarily stored incoming raw sensor text data.

7. **Job Flow / Pipeline Stages:**

- Establish connection with an IoT sensor.
- Continuously receive raw text data packets from the connected sensor.
- Buffer or temporarily store the incoming raw sensor text.

8. Data Transformations / Business Logic:

- Step: Data Reception
- Description: Handle incoming raw text data streams.
- Transformation Logic: Direct reception and queuing to internal buffer.

9. Error Handling and Logging:

- Error Handling: Not explicitly detailed in FRD, but implies handling for connection stability and data reception.
- Logging: Not explicitly detailed in FRD for this stage.
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TR-GW-002: Clean and Standardize Sensor Data

1. Technical Requirement ID: TR-GW-002

2. Related Functional Requirement(s): FR-GW-002

3. **Objective:** Implement data cleaning algorithms to mitigate noisy data points and transform the cleaned raw sensor data into the company's predefined standard schema.

4. Target Cluster Configuration:

- Cluster Name: Not specified in FRD
- Databricks Runtime Version: Not specified in FRD
- Node Type: Not specified in FRD
- Driver Node: Not specified in FRD
- Worker Nodes: Not specified in FRD
- Autoscaling: Not specified in FRD
- Auto Termination: Not specified in FRD
- Libraries Installed: Not specified in FRD

5. Source Data Details:

- Dataset Name: Ingested Raw Sensor Data
- Location (Path/Table): Gateway's internal buffer (from TR-GW-001)
- Format: Raw text data
- Description: Raw sensor data successfully ingested by the gateway.

6. Target Data Details:

- Output Name: Normalized, Schema-Compliant Sensor Data
- Location: Gateway's internal output for further processing
- Format: Company's standard data schema (details not specified)
- Description: Cleaned and standardized sensor data, compliant with the company's predefined schema.

7. Job Flow / Pipeline Stages:

- Receive a block of raw sensor text data.
- Apply data cleaning algorithms to identify and rectify or filter out "noisy" data points.
- Identify the specific data structure of the cleaned raw sensor data.
- Transform cleaned data elements according to predefined mapping rules.
- Structure the transformed data into the company's standard schema.
- Output the normalized, schema-compliant data for further processing.

8. Data Transformations / Business Logic:

- Step: Data Cleaning
- Description: Mitigate "noisy" data points.
- Transformation Logic: Apply smoothing filters for location data; discard outlier readings based on configured rules.
- Step: Schema Identification
- Description: Determine the sensor's original data structure.
- Transformation Logic: Based on sensor ID or manufacturer metadata.
- Step: Data Standardization
- Description: Convert cleaned data elements to the standard schema.
- Transformation Logic: Apply predefined mapping rules for the sensor type to structure data into the company's standard schema.

9. Error Handling and Logging:

- Error Handling: Implementation of cleaning algorithms to rectify/filter noisy data points.
- Logging: Not explicitly detailed in FRD for this stage.
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TR-GW-003: Route Engine Alerts to Maintenance Agent

1. **Technical Requirement ID:** TR-GW-003

2. **Related Functional Requirement(s):** FR-GW-003

3. **Objective:** Implement logic to detect "Engine Alerts" from normalized sensor data, extract relevant details, format them, and dispatch these alerts to the designated Maintenance Agent, logging the dispatch.

4. **Target Cluster Configuration:**

- Cluster Name: Not specified in FRD
- Databricks Runtime Version: Not specified in FRD
- Node Type: Not specified in FRD
- Driver Node: Not specified in FRD
- Worker Nodes: Not specified in FRD

- Autoscaling: Not specified in FRD
- Auto Termination: Not specified in FRD
- Libraries Installed: Not specified in FRD

5. **Source Data Details:**

- Dataset Name: Normalized Sensor Data
- Location (Path/Table): Gateway's internal output (from TR-GW-002)
- Format: Standard schema
- Description: Normalized sensor data containing indicators or values for "Engine Alert" classification.

6. **Target Data Details:**

- Output Name: Formatted Engine Alert Message
- Location: Configured communication channel (e.g., API endpoint, message queue) to Maintenance Agent
- Format: Suitable for Maintenance Agent's system (details not specified)
- Description: Contains truck ID, engine parameter, timestamp, severity.
- Output Name: Alert Dispatch Log
- Location: Gateway's internal log
- Format: Log entries
- Description: Records the successful dispatch of "Engine Alert" messages.

7. **Job Flow / Pipeline Stages:**

- Receive a batch of normalized sensor data.
- Process data to detect predefined "Engine Alert" conditions or flags.
- Extract relevant details (e.g., truck ID, engine parameter, timestamp, severity).
- Format extracted alert details into a message suitable for the Maintenance Agent's system.
- Dispatch the formatted "Engine Alert" message via the configured communication channel.
- Log the successful dispatch of the alert.

8. **Data Transformations / Business Logic:**

- Step: Alert Condition Detection
- Description: Identify engine alert conditions.
- Transformation Logic: Evaluate normalized data against predefined "Engine Alert" conditions or flags.
- Step: Alert Data Extraction
- Description: Isolate and extract specific alert parameters.
- Transformation Logic: Extract truck ID, engine parameter, timestamp, severity from the alert-triggering data.
- Step: Message Formatting

- Description: Prepare alert data for dispatch.
- Transformation Logic: Format extracted details into a message structure compatible with the Maintenance Agent's system.

9. Error Handling and Logging:

- Error Handling: Not explicitly detailed for dispatch failure.
- Logging: Log successful dispatch of "Engine Alert" messages.
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TR-GW-004: Route Location Updates to Customer Tracking Agent

1. Technical Requirement ID: TR-GW-004

2. Related Functional Requirement(s): FR-GW-004

3. **Objective:** Implement logic to identify "Location Updates" from normalized sensor data, extract relevant details, format them, and dispatch these updates to the designated Customer Tracking Agent, logging the dispatch.

4. Target Cluster Configuration:

- Cluster Name: Not specified in FRD
- Databricks Runtime Version: Not specified in FRD
- Node Type: Not specified in FRD
- Driver Node: Not specified in FRD
- Worker Nodes: Not specified in FRD
- Autoscaling: Not specified in FRD
- Auto Termination: Not specified in FRD
- Libraries Installed: Not specified in FRD

5. Source Data Details:

- Dataset Name: Normalized Sensor Data
- Location (Path/Table): Gateway's internal output (from TR-GW-002)
- Format: Standard schema
- Description: Normalized sensor data containing current location information.

6. Target Data Details:

- Output Name: Formatted Location Update Message
- Location: Configured communication channel to Customer Tracking Agent
- Format: Suitable for Customer Tracking Agent's system (details not specified)
- Description: Contains truck ID, latitude, longitude, timestamp.
- Output Name: Location Update Dispatch Log
- Location: Gateway's internal log

- Format: Log entries
- Description: Records the successful dispatch of "Location Update" messages.

7. Job Flow / Pipeline Stages:

- Receive a batch of normalized sensor data.
- Process data to identify current "Location Update" information (e.g., GPS coordinates, speed, direction).
- Extract all relevant location details (e.g., truck ID, latitude, longitude, timestamp).
- Format extracted location details into a message suitable for the Customer Tracking Agent's system.
- Dispatch the formatted "Location Update" message via the configured communication channel.
- Log the successful dispatch of the location update.

8. Data Transformations / Business Logic:

- Step: Location Data Identification
- Description: Identify current location update information.
- Transformation Logic: Detect and parse GPS coordinates, speed, direction, etc., from normalized data.
- Step: Location Data Extraction
- Description: Isolate and extract specific location parameters.
- Transformation Logic: Extract truck ID, latitude, longitude, and timestamp from the location data.
- Step: Message Formatting
- Description: Prepare location data for dispatch.
- Transformation Logic: Format extracted details into a message structure compatible with the Customer Tracking Agent's system.

9. Error Handling and Logging:

- Error Handling: Not explicitly detailed for dispatch failure.
- Logging: Log successful dispatch of location updates.